

Problem 17. Find all triangular numbers which are also perfect squares.

(Note: A triangular number is a number of the form $1 + 2 + \cdots + n$.)

Solution. We wish to find all positive integers m and n such that $m^2 = 1 + 2 + \cdots + n = \frac{n(n+1)}{2}$. This is the same as

$$(2n+1)^2 - 8m^2 = 1. \quad (1)$$

Notice that $(x, y) = (3, 1)$ is the minimal solution to Pell's equation $x^2 - 8y^2 = 1$. Hence all solutions (x_k, y_k) are given by $x_k + y_k\sqrt{8} = (3 + \sqrt{8})^k$. Since $x_k - y_k\sqrt{8} = (3 - \sqrt{8})^k$, we have

$$x_k = \frac{(3 + \sqrt{8})^k + (3 - \sqrt{8})^k}{2}.$$

Clearly, from $x^2 - 8y^2 = 1$, all solutions x_k must be odd. As $n \geq 1$, we need $x \geq 3$. So each x_k with $k \geq 1$ corresponds to a solution to (1). Thus, the required triangular numbers are

$$\frac{n(n+1)}{2} = \frac{(x_k-1)(x_k+1)}{8} = \frac{1}{8} \left(\frac{((3 + \sqrt{8})^k + (3 - \sqrt{8})^k)^2}{4} - 1 \right).$$

□

Problem 18 (IMO Shortlist 2000 N5). Prove that there exist infinitely many positive integers n such that $p = nr$, where p and r are respectively the semiperimeter and the inradius of a triangle with integer side lengths.

Solution. Let the side lengths of the triangle be $a + b$, $b + c$ and $c + a$ respectively. Using the area formulae, we have $pr = \sqrt{pabc}$, and so

$$\frac{p}{r} = \frac{p^2}{\sqrt{pabc}}.$$

The equation $p = nr$ is the same as $n^2abc = p^3 = (a + b + c)^3$. If we can show that there are infinitely many positive integers n such that

$$n^2abc = (a + b + c)^3$$

has a positive integer solution, then $a + b$, $b + c$ and $c + a$ must be positive integers satisfying the triangle inequality, and we are done.

Observe that $(x, y) = (1, 1)$ is a solution to the generalized Pell's equation $x^2 - 5y^2 = -4$. Therefore the equation admits infinitely many solutions. Given a solution (x, y) to the equation, let

$$a = \frac{3y + x}{2}, \quad b = \frac{3y - x}{2}, \quad c = (ab - 1)(a + b).$$

Note that a, b, c are positive integers since x and y have the same parity and $x < \sqrt{5}y < 3y$. Now,

$$\frac{(a + b + c)^3}{abc} = \frac{a^3b^3(a + b)^3}{ab(ab - 1)(a + b)} = \frac{a^2b^2(a + b)^2}{ab - 1}.$$

Also, we have $ab - 1 = \frac{9y^2 - x^2}{4} - 1 = y^2$ and $a + b = 3y$. Therefore,

$$\frac{(a + b + c)^3}{abc} = 9a^2b^2 = (3ab)^2.$$

So we can take $n = 3ab$. As there are infinitely many pairs of (x, y) , there are infinitely many pairs of (a, b) , and hence infinitely many n satisfying the condition. □

Problem 19. A *fissile square* is a positive integer which is a perfect square, and whose digits form two perfect squares in a row, which will be called the *left square* and the *right square*. For example, 49, 1444, and 1681 are fissile squares (decomposing as $4 \mid 9$, $144 \mid 4$, and $16 \mid 81$ respectively). Neither the left nor the right square may begin with the digit 0. Which perfect squares are the right square of infinitely many fissile squares?

Solution. The answer is all perfect squares that have an odd number of digits.

A square n^2 is a right square infinitely often if and only if

$$10^m y^2 + n^2 = x^2 \quad (1)$$

has infinitely many positive integer solutions (x, y) , where m is the number of digits of n^2 . This is because each such solution corresponds to a fissile square x^2 with left square y^2 and right square n^2 .

If m is even, by letting $m = 2k$, we have

$$n^2 = (x - 10^k y)(x + 10^k y).$$

Each factor is at most n^2 , so their sum is at most $2n^2$. Hence $x < n^2$, so it can only take finitely many values, which gives a finite number of solutions.

If m is odd, then 10^m is not a perfect square, so there are infinitely many solutions to the following Pell's equation

$$a^2 - 10^m b^2 = 1.$$

For each such solution (a, b) , we can generate a new solution $(x, y) = (na, nb)$ to (1). So we are done. $\square$

Problem 20 (US Math Competition Association 2019 Premier). Find all pairs of positive integers (m, n) such that $(2^m - 1)(2^n - 1)$ is a perfect square.

Solution. The solutions are $m = n$ or $\{m, n\} = \{3, 6\}$. These obviously work.

Let $d = \gcd(m, n)$. It is well known that $\gcd(2^m - 1, 2^n - 1) = 2^d - 1$. Note that

$$(2^m - 1)(2^n - 1) = (2^d - 1)^2 \cdot \frac{2^m - 1}{2^d - 1} \cdot \frac{2^n - 1}{2^d - 1}.$$

Since the two fractions are relatively prime, if $(2^m - 1)(2^n - 1)$ is a perfect square, then each of $\frac{2^m - 1}{2^d - 1}$ and $\frac{2^n - 1}{2^d - 1}$ is a perfect square.

If m and d have different parities, then m must be even and d must be odd (since $d \mid m$). Let $m = 2k$ where $d \mid k$. Then

$$\frac{2^m - 1}{2^d - 1} = \frac{2^k - 1}{2^d - 1} \cdot (2^k + 1).$$

Since $\gcd(2^k - 1, 2^k + 1) = 1$, the integers $\frac{2^k - 1}{2^d - 1}$ and $2^k + 1$ are relatively prime. Thus, each of them is a perfect square. In particular, $2^k + 1 = \ell^2$ for some positive integer ℓ . This yields $2^k = (\ell - 1)(\ell + 1)$, and so $\ell - 1$ and $\ell + 1$ are powers of 2. The only possibility is $\ell = 3$, so $k = 3$, $m = 6$ and $d = 3$.

If m and d have the same parity, let $m = 2a + d$ for some nonnegative integer a . Let $\frac{2^{2a+d} - 1}{2^d - 1} = b^2$. This can be rewritten as

$$2^d(2^a)^2 - (2^d - 1)b^2 = 1. \quad (1)$$

We use the substitution

$$x = -(2^d - 1)b + 2^{d+a}, \quad y = b - 2^a.$$

Equivalently, these mean $2^a = x + (2^d - 1)y$ and $b = x + 2^d y$. Putting these into (1) and simplifying, we obtain

$$x^2 - 2^d(2^d - 1)y^2 = 1.$$

The coefficient $D = 2^d(2^d - 1)$ is a square if and only if both 2^d and $2^d - 1$ are squares, which is impossible. So this is Pell's equation. When $y = 1$, $x^2 = 2^{2d} - 2^d + 1$, which gives $(2^d - 1)^2 < x^2 < (2^d)^2$. This is a contradiction. When $y = 2$,

$$x^2 = 2^{2d+2} - 2^{d+2} + 1 = (2^{d+1} - 1)^2.$$

Thus, $(x, y) = (2^{d+1} - 1, 2)$ is the fundamental solution. The general solution (x_j, y_j) is given by $x_j + y_j \sqrt{D} = (2^{d+1} - 1 + 2\sqrt{D})^j$. By induction, it is easy to see that x_j is odd and y_j is even (which also holds if x, y are the negatives of some x_j, y_j , and also for the solution $(x, y) = (1, 0)$). But then $2^a = x + (2^d - 1)y$ must be odd, so we must have $a = 0$. It follows that $m = d$.

The same results hold for n . Therefore, one of the following cases must hold: $m = d = n$; $m = 6, n = d = 3$; $n = 6, m = d = 3$. $\square$